Supplemental Digital Content 2 – Table with overview of collected data

First author	Year of publication	Definition used	Secondary definition used	Tertiary definition used	Surgical specialty	Mean or median age in years	Preoperative opioid status	Outcome type	Prevalence of prolonged use	Method of opioid use	OMEs	Risk factors found*
Zhang, Y.	2020	12 months			Spine	50-59.9	Naïve	Primary	4.70%	Oral	Reported	Preoperative Opioid Use
Wells, B.	2020	9 months			Urologic	30-39.9	Naïve	Primary	1.60%	Oral	Reported	None reported
Harris, A.B.	2020	3 months			Spine	50-59.9	Combined	Primary	Not reported	Oral	Not reported	Mood Disorder, Anxiety
Harbaugh, C.M.	2019	3 months			Mixed	<20	Naïve	Primary	2.50%	Oral	Reported	None reported
Bicket, M.C.	2019	3 months			Mixed	40-49.9	Naïve	Primary	10.20%	Oral	Reported	Preoperative Opioid Use, Tobacco, Pain or Chronic Pain, Anxiety, Mood Disorder, History of Substance Use, Female Sex
Lasiello-Rangeley, J.	2020	3 months			Orthopedic	Not reported	Naïve	Primary	4%	Oral	Reported	Female Sex, Medicare/Medicaid, Mood disorder
Xu, Y.	2020	2 months	3 months	Not time-based	Surgical Oncology	60-69.9	Naïve	Primary	8.60%	Oral	Reported	Preoperative Opioid Use
Sabesan, V.J.	2019	3 months			Orthopedic	50-59.9	Combined	Primary	46.50%	Oral	Not reported	Preoperative Opioid Use
Anderson, A.B.	2020	3 months			Orthopedic	20-29.9	Combined	Secondary	Not reported	Oral	Not reported	None reported
Wright, A.K.	2020	Not time-based			Spine	50-59.9	Combined	Primary	19.80%	Oral	Not reported	Preoperative Opioid Use, Mood Disorder, Benzodiazepines
Anderson, J.T.	2015	1.5 months			Spine	40-49.9	Combined	Primary	57.40%	Oral	Reported	Preoperative Opioid Use, Mood Disorder
Hoji, S.	2019	3 months			Cardiac	60-69.9	Combined	Primary	21.70%	Oral	Not reported	None reported
Harris, A.B.	2020	3 months			Spine	50-59.9	Combined	Primary	17%	Oral	Not reported	Preoperative Opioid Use, History of Substance Use, Mood Disorder, Anxiety, Tobacco, Treatment Outside Northeast US
Peng, J.	2017	3 months			Otolaryngology	60-69.9	Combined	Primary	41%	Oral	Not reported	Preoperative Opioid Use, Tobacco
Wen, X.	2020	6 months			Gynecologic	Not reported	Naïve	Primary	1.28%	Oral	Reported	Tobacco, Pain or Chronic Pain
Thiels, C.A.	2019	3 months	6 months	Not time-based	Mixed	Not reported	Naïve	Primary	7.07%	Oral	Reported	None reported
Swenson, C.W.	2018	Not time-based			Gynecologic	Not reported	Naïve	Primary	0.50%	Oral	Reported	Preoperative Opioid Use, Mood Disorder, Anxiety
Chaudhary, M.A.	2019	Not time-based			Mixed	40-49.9	Combined	Primary	7.40%	Oral	Not reported	Preoperative Opioid Use
Katakam, A.	2020	1 month	3 months		Orthopedic	60-69.9	Combined	Primary	9.00%	Oral	Not reported	Preoperative Opioid Use, Benzodiazepines
Nicholson, A.D.	2020	1.5 months			Orthopedic	50-59.9	Combined	Primary	28%	Oral	Reported	Preoperative Opioid Use, Medicare/Medicaid
Karhade, A.V.	2019	3 months			Spine	40-49.9	Combined	Primary	7.70%	Oral	Not reported	Preoperative Opioid Use, Mood Disorder, Tobacco, Benzodiazepines, History of Substance Use
Karhade, A.V.	2019	3 months			Orthopedic	60-69.9	Combined	Primary	6.30%	Oral	Not reported	None reported
Ly, D.P.	2019	6 months			Orthopedic	70-79.9	Naïve	Primary	0.30%	Oral	Reported	None reported
Agnew alla, A.	2020	3 months			Orthopedic	Not reported	Combined	Primary	29.90%	Oral	Not reported	Preoperative Opioid Use, Mood Disorder, Pain or Chronic Pain, Tobacco
Neuman, M.D.	2020	3 months			Mixed	50-59.9	Naïve	Primary	11.40%	Oral	Reported	None reported
Reid, D.B.C.	2019	1 month			Spine	50-59.9	Combined	Secondary	33.30%	Oral	Reported	Preoperative Opioid Use
Smith, M.E.	2019	12 months			General	40-49.9	Combined	Secondary	6.30%	Other	Not reported	Tobacco, Mood Disorder
Leroux, T.S.	2019	3 months			Orthopedic	Not reported	Naïve	Primary	14.60%	Oral	Not reported	Mood Disorder, Anxiety, Female Sex
Wang, M.C.	2020	12 months			Spine	50-59.9	Combined	Primary	32.00%	Oral	Not reported	Preoperative Opioid Use, Tobacco
Brescia, A.A.	2019	3 months			Surgical Oncology	60-69.9	Naïve	Primary	14%	Oral	Not reported	Male Sex
Sharma, M.	2018	3 months	Not time-based	Not time-based	Spine	60-69.9	Combined	Primary	9.90%	Oral	Not reported	Preoperative Opioid Use
Oy, C.J.	2019	3 months			Orthopedic	40-49.9	Combined	Primary	27.70%	Oral	Not reported	Preoperative Opioid Use, Pain or Chronic Pain, History of Substance Use, Tobacco, Anxiety
Brummett, C.M.	2019	3 months			Mixed	Not reported	Naïve	Secondary	Not reported	Oral	Not reported	None reported
Lee, J.S.	2020	3 months			Mixed	40-49.9	Naïve	Secondary	6%	Oral	Not reported	Preoperative Opioid Use, Mood Disorder, Anxiety, Female Sex, Tobacco, Pain or Chronic Pain
Sharma, M.	2018	Not time-based	Not time-based	Not time-based	Spine	60-69.9	Combined	Secondary	9.90%	Oral	Not reported	None reported
Delaney, L.D.	2020	3 months			Orthopedic	Not reported	Naïve	Primary	6.90%	Oral	Reported	None reported
Chettia, K.	2020	3 months			Orthopedic	60-69.9	Combined	Secondary	32%	Oral	Reported	Preoperative Opioid Use
Woente, M.R.	2020	3 months			General	50-59.9	Combined	Primary	44.30%	Oral	Reported	Preoperative Opioid Use, Mood Disorder
Chen, Q.	2019	3 months			Orthopedic	60-69.9	Combined	Secondary	19.60%	Oral	Not reported	None reported
Kim, Y.	2018	3 months			General	40-49.9	Combined	Secondary	28.90%	Oral	Reported	None reported
Brescia, A.A.	2019	3 months			Mixed	70-79.9	Naïve	Primary	12.80%	Oral	Reported	Preoperative Opioid Use, History of Substance Use, Female Sex, Medicare/Medicaid, Pain or Chronic Pain, Tobacco
Eley, N.	2020	3 months			Spine	Not reported	Naïve	Secondary	2.80%	Oral	Reported	None reported
Jafari, A.	2018	2 months			Otolaryngology	50-59.9	Combined	Secondary	18%	Oral	Not reported	Preoperative Opioid Use, Tobacco
Dunn, L.K.	2018	12 months			Spine	50-59.9	Combined	Primary	42.50%	Oral	Reported	Preoperative Opioid Use, Pain or Chronic Pain
Ward, A.	2020	3 months	3 months		Mixed	Not reported	Naïve	Primary	2.00%	Oral	Not reported	Mood Disorder, Female Sex, Pain or Chronic Pain
Sun, E.C.	2016	3 months			Mixed	40-49.9	Naïve	Primary	0.60%	Other	Not reported	Male Sex, Benzodiazepines, Mood Disorder, History of Substance Use
Li, N.Y.	2020	3 months			Spine	Not reported	Combined	Secondary	23%	Oral	Not reported	None reported
Wilson, L.A.	2020	3 months			Orthopedic	Not reported	Unknown	Secondary	Not reported	Oral	Not reported	None reported
Sun, E.C.	2017	3 months			Orthopedic	50-59.9	Combined	Primary	9.80%	Other	Not reported	None reported
Alam, A.	2012	2 months			Mixed	70-79.9	Naïve	Primary	10.30%	Other	Not reported	Preoperative Opioid Use
Maciejewski, M.L.	2020	Not time-based			General	Not reported	Combined	Primary	Not reported	Oral	Reported	None reported
Bartels, K.	2018	1 month			Mixed	Not reported	Combined	Primary	43.60%	Oral	Not reported	Preoperative Opioid Use
Karhade, A.V.	2019	3 months			Spine	50-59.9	Combined	Primary	9.90%	Oral	Not reported	Preoperative Opioid Use, Tobacco, Medicare/Medicaid
Reid, D.B.C.	2019	1 month			Orthopedic	50-59.9	Combined	Secondary	25.90%	Oral	Reported	Preoperative Opioid Use, Benzodiazepines
Berger, J.	2019	3 months	Not time-based		Urologic	Not reported	Naïve	Primary	6.20%	Oral	Reported	None reported
Orfiel, N.J.	2020	3 months			Orthopedic	40-49.9	Naïve	Primary	5.30%	Oral	Not reported	Medicare/Medicaid, Female Sex
Clement, K.C.	2020	3 months			Cardiac	50-59.9	Naïve	Primary	5.50%	Oral	Reported	Tobacco
Brummett, C.M.	2017	3 months			Mixed	Not reported	Naïve	Primary	6.01%	Oral	Reported	Preoperative Opioid Use, Anxiety, Tobacco, Pain or Chronic Pain, History of Substance Use, Treatment Outside Northeast US
Tam, C.A.	2019	3 months			Urologic	Not reported	Combined	Primary	8.10%	Oral	Reported	Mood Disorder, Pain or Chronic Pain
Santoso, K.B.	2020	3 months			Mixed	Not reported	Naïve	Primary	9.80%	Oral	Not reported	Preoperative Opioid Use, Mood Disorder, Pain or Chronic Pain, Medicare/Medicaid, History of Substance Use
Lee, J.S.J.	2017	1 month	3 months		Surgical Oncology	50-59.9	Combined	Primary	10.40%	Oral	Not reported	Preoperative Opioid Use
Finney, F.T.	2019	3 months			Orthopedic	40-49.9	Naïve	Primary	6.20%	Oral	Reported	Preoperative Opioid Use, Treatment Outside Northeast US, History of Substance Use, Mood Disorder, Anxiety, Pain or Chronic Pain
Kang, C.	2019	2 months			Urologic	50-59.9	Combined	Primary	6.70%	Oral	Reported	None reported
Sharma, M.	2020	Not time-based	Not time-based	Not time-based	Spine	50-59.9	Combined	Primary	28.34%	Oral	Not reported	Preoperative Opioid Use
Ren, M.	2020	24 months			Spine	60-69.9	Combined	Primary	39%	Oral	Reported	Preoperative Opioid Use, Mood Disorder, Anxiety
Chen, M.J.	2019	6 months			Orthopedic	Not reported	Naïve	Primary	2.60%	Oral	Not reported	None reported
Tran, T.	2020	3 months			Orthopedic	60-69.9	Combined	Secondary	5.30%	Other	Reported	None reported
Pugely, A.J.	2018	12 months			Spine	60-69.9	Combined	Primary	28.30%	Oral	Not reported	Preoperative Opioid Use, Mood Disorder, Anxiety, Pain or Chronic Pain, History of Substance Use
Khazi, Z.M.	2019	3 months			Orthopedic	Not reported	Combined	Primary	12.40%	Oral	Not reported	Preoperative Opioid Use, Tobacco
Kalakoti, P.	2018	12 months			Spine	Not reported	Combined	Secondary	28.30%	Other	Not reported	Preoperative Opioid Use, Mood Disorder, Anxiety, Pain or Chronic Pain, Male Sex, History of Substance Use
Nguyen, A.V.	2020	12 months			Spine	Not reported	Combined	Primary	6%	Oral	Not reported	Preoperative Opioid Use, Anxiety, Medicare/Medicaid, Benzodiazepines
Reid, D.B.C.	2020	3 months			Spine	50-59.9	Combined	Secondary	22.75%	Oral	Reported	Preoperative Opioid Use
Reid, D.B.C.	2020	1 month			Orthopedic	50-59.9	Combined	Secondary	Not reported	Oral	Reported	Preoperative Opioid Use, Male Sex
Labarian, L.	2020	3 months			Spine	Not reported	Combined	Primary	79.60%	Oral	Not reported	Male Sex
Anciano Granadillo, V.	2018	3 months			Orthopedic	Not reported	Combined	Primary	27.40%	Oral	Not reported	Preoperative Opioid Use, History of Substance Use, Mood Disorder, Tobacco, Male Sex, Pain or Chronic Pain
Young, J.C.	2020	3 months			Gynecologic	40-49.9	Naïve	Primary	6.18%	Oral	Not reported	None reported
Bennett, K.G.	2018	3 months			Plastic	<20	Naïve	Primary	4.40%	Oral	Not reported	Treatment Outside Northeast US
Harbaugh, C.M.	2018	3 months			Pediatric	Not reported	Naïve	Primary	4.80%	Oral	Not reported	Preoperative Opioid Use, History of Substance Use, Female Sex, Pain or Chronic Pain
Nelson, D.B.	2020	3 months			Surgical Oncology	Not reported	Combined	Primary	31%	Oral	Not reported	None reported
Bennett, K.G.	2019	3 months			Plastic	40-49.9	Naïve	Primary	6.10%	Oral	Reported	Mood Disorder, Pain or Chronic Pain, Tobacco, Anxiety
Nelson, D.B.	2019	3 months			Surgical Oncology	Not reported	Combined	Secondary	27.00%	Oral	Not reported	None reported
Saraswathula, A.	2019	3 months			Otolaryngology	70-79.9	Combined	Primary	33.30%	Oral	Not reported	Preoperative Opioid Use
Brock, J.L.	2019	12 months			Orthopedic	Not reported	Combined	Primary	9.40%	Oral	Not reported	Preoperative Opioid Use, Mood Disorder, History of Substance Use
Rogero, R.	2019	3 months			Orthopedic	50-59.9	Naïve	Primary	1.50%	Oral	Reported	None reported
Young, J.C.	2019	3 months			Mixed	40-49.9	Naïve	Primary	8%	Oral	Reported	None reported
Karhade, A.V.	2020	3 months	Not time-based		Spine	60-69.9	Naïve	Primary	4.10%	Oral	Not reported	Tobacco, Benzodiazepines
Hockley, A.	2020	6 months			Spine	60-69.9	Combined	Primary	19.60%	Oral	Reported	Preoperative Opioid Use, Pain or Chronic Pain
Connolly, J.	2017	24 months			Spine	40-49.9	Combined	Primary	29.34%	Oral	Not reported	Preoperative Opioid Use, Pain or Chronic Pain
Clement, K.C.	2020	3 months			Cardiac	50-59.9	Naïve	Primary	8.10%	Oral	Reported	History of Substance Use, Anxiety, Tobacco, Female Sex
Fields, A.C.	2019	3 months			General	Not reported	Combined	Primary	10.60%	Oral	Reported	None reported
Jain, N.	2018	1.5 months			Spine	60-69.9	Combined	Primary	13.80%	Oral	Not reported	Preoperative Opioid Use, Mood Disorder, Anxiety, Female Sex, Tobacco, Medicare/Medicaid, History of Substance Use
Kim, K.Y.	2018	6 months			Orthopedic	60-69.9	Non-naïve	Primary	43.40%	Oral	Not reported	Male Sex, Tobacco
Hozack, B.A.	2019	3 months			Orthopedic	50-59.9	Combined	Secondary	Not reported	Oral	Reported	None reported
Rojas, E.O.	2020	3 months			Orthopedic	Not reported	Combined	Primary	14.14%	Oral	Not reported	Preoperative Opioid Use
Khazi, Z.M.	2020	3 months			Orthopedic	Not reported	Combined	Primary	19.90%	Oral	Not reported	Preoperative Opioid Use, Mood Disorder, Anxiety, Male Sex, Tobacco
Khazi, Z.M.	2020	1.5 months			Orthopedic	20-29.9	Combined	Primary	33.30%	Oral	Not reported	Preoperative Opioid Use
Bell, J.E.	2020	3 months			Orthopedic	Not reported	Combined	Secondary	48.90%	Oral	Not reported	None reported
Prentice, H.A.	2019	3 months			Orthopedic	60-69.9	Combined	Secondary	27.90%	Oral	Not reported	History of Substance Use, Female Sex, Pain or Chronic Pain, Mood Disorder, Anxiety
Rasilico, JM.	2019	3 months			Orthopedic	50-59.9	Naïve	Primary	20.65%	Oral	Reported	None reported
Marcuzo, D.F.	2017	3 months			Plastic	Not reported	Naïve	Primary	10%	Oral	Reported	Anxiety
Cron, D.C.	2021	Not time-based	Not time-based	Not time-based	General	50-59.9	Combined	Secondary	14.10%	Oral	Not reported	Preoperative Opioid Use
Stark, N.	2017	3 months			Mixed	60-69.9	Naïve	Primary	10.50%	Oral	Reported	Pain or Chronic Pain
Wu, L.	2020	3 months			Orthopedic	Not reported	Combined	Primary	12%	Oral	Not reported	Preoperative Opioid Use, History of Substance Use, Mood Disorder, Anxiety, Tobacco, Pain or Chronic Pain, Benzodiazepines
Rozet, I.	2016	3 months			Orthopedic	30-39.9	Combined	Primary	30%	Oral	Not reported	Preoperative Opioid Use, Tobacco
Shen, C.	2020	3 months	6 months		Surgical Oncology	Not reported	Naïve	Primary	18.06%	Oral	Reported	History of Substance Use, Mood Disorder, Anxiety
Clarke, H.	2014	3 months			Mixed	Not reported	Naïve	Primary	3.10%	Other	Not reported	Benzodiazepines
Inacio, M.C.S.	2016	3 months	4 months		Orthopedic	>80	Combined	Primary	5.20%	Oral	Reported	Mood Disorder, Pain or Chronic Pain, Female Sex
Reyes, A.A.	2020	Not time-based			Spine	50-59.9	Combined	Primary	50%	Oral	Reported	Preoperative Opioid Use
Schoenfeld, A.J.	2017	Not time-based			Spine	40-49.9	Naïve	Primary	0.02%	Oral	Not reported	Mood Disorder
Yang, S.	2019	1.5 months			Spine	Not reported	Combined	Primary	9.78%	Oral	Not reported	Preoperative Opioid Use, Anxiety, Female Sex
Hadlandsmyth, K.	2018	12 months			Orthopedic	60-69.9	Combined	Primary	13%	Oral	Not reported	Preoperative Opioid Use, History of Substance Use
Zaveri, S.	2020	3 months			Mixed	Not reported	Combined	Primary	5%	Oral	Not reported	Female Sex
Gil, J.A.	2019	3 months			Mixed	Not reported	Naïve	Primary	8.30%	Oral	Reported	Mood Disorder, Anxiety, Female Sex, Pain or Chronic Pain
Johnson, S.P.	2016	3 months			Orthopedic	Not reported	Naïve	Primary	12.90%	Oral	Reported	Female Sex, Tobacco
Shkolyar, E.	2020	3 months	6 months		Urologic	Not reported	Naïve	Primary	6.90%	Oral	Reported	Tobacco
Barham, D.W.	2019	3 months			Urologic	30-39.9	Naïve	Secondary	4.30%	Oral	Reported	None reported
Adogwa, O.	2019	12 months			Spine	Not reported	Combined	Secondary	66.00%	Oral	Not reported	Preoperative Opioid Use, Female Sex
Tyrag, D.S.	2020	3 months			Spine	Not reported	Unknown	Secondary	13.20%	Oral	Not reported	None reported
Katzman, C.	2020	3 months			Mixed	Not reported	Combined	Primary	11.70%	Oral	Not reported	Preoperative Opioid Use, History of Substance Use, Mood Disorder, Tobacco, Pain or Chronic Pain, Benzodiazepines
Steen, T.	2020	3 months			Orthopedic	70-79.9	Combined	Primary	31.80%	Oral	Not reported	Benzodiazepines
Grace, T.R.	2018	1.5 months			Orthopedic	60-69.9	Combined	Secondary	33.50%	Other	Reported	Preoperative Opioid Use
Kueh, M.P.	2018	3 months			Mixed	40-49.9	Naïve	Secondary	Not reported	Oral	Not reported	None reported
Goesling, J.	2016	6 months			Orthopedic	60-69.9	Combined	Primary	15.35%	Oral	Reported	Preoperative Opioid Use, Pain or Chronic Pain
Politzer, C.S.	2018	6 months			Orthopedic	Not reported	Combined	Primary	34.80%	Other	Not reported	Preoperative Opioid Use, Female Sex
Tan, T.L.	2020	6 months			Orthopedic	Not reported	Combined	Primary	10.50%	Oral	Reported	Preoperative Opioid Use, History of Substance Use
Kuo, J.H.	2020	3 months			General	50-59.9	Naïve	Primary	7.40%	Oral	Reported	Mood Disorder, Anxiety, Medicare/Medicaid, Treatment Outside Northeast US, Pain or Chronic Pain, History of Substance Use
Wright, J.D.	2019	3 months			Gynecologic	40-49.9	Naïve	Primary	6.80%	Oral	Reported	Treatment Outside Northeast US, Mood Disorder, Anxiety, History of Substance Use, Medicare/Medicaid, Female Sex
Deyo, R.A.	2018	Not time-based			Spine	Not reported	Combined	Primary	43.92%	Oral	Reported	Preoperative Opioid Use, Benzodiazepines
Karhade, A.V.	2019	3 months			Spine	40-49.9	Combined	Primary	8.60%	Oral	Not reported	None reported
Stafford, C.	2018	1 month			General	60-69.9	Combined	Primary	4%	Oral	Not reported	Pain or Chronic Pain, Benzodiazepines